CHANDIGARH ENGINEERING COLLEGE JHANJERI, MOHALI
Department of Computer Science and Engineering
Affiliated to I.K. Gujral Punjab Technical University, Jalandhar (Batch: 2023–2027)

PROJECT SYNOPSIS
EDGE AI SMART PROSTHETIC HAND WITH ADAPTIVE GRIP LEARNING

Project Nature: Engineering Clinics / Project-II (Semester VII)
Academic Year: 2026–2027
Domain: Edge AI, Biomechatronics, Cyber-Physical Simulation

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> **System Scope & Academic Integrity Disclaimer:** This synopsis documents a verified **software simulation prototype** comprising a FastAPI backend and a React 19 / TypeScript frontend. All sEMG waveforms, distal FSR contact pressures, and shear slip disturbances are digitally simulated. Performance metrics (100% accuracy, 1.0000 Macro-F1) reflect synthetic dataset separability, while latencies represent host CPU execution. Physical electrodes, servomotors, and clinical amputee trials represent a future 5-phase hardware deployment roadmap.

1. Introduction & Clinical Motivation

Upper-limb loss severely impairs manual manipulation, occupational capacity, and daily independence. While commercial myoelectric prostheses are clinically available, over 80% utilize open-loop dual-threshold switching. In this paradigm, muscle contractions actuate motors at fixed speeds until mechanical stall occurs, completely devoid of tactile awareness. Consequently, amputees suffer high rates of payload slippage and crush damage, while experiencing extreme cognitive fatigue due to constant visual supervision. Conversely, academic deep learning approaches require multi-megabyte footprints and cloud tethering that introduce latency jitter (150–650 ms) and privacy concerns.

This project resolves this dual challenge by developing an **Edge AI Smart Prosthetic Hand** that decouples high-level intention classification from low-level closed-loop tactile force regulation, executing within a sub-2 ms host latency profile suitable for embedded microcontrollers.

2. Formal Problem Statement & Quantifiable Objectives

Problem Statement: Existing myoelectric prostheses lack low-latency, embedded-compatible pattern recognition combined with autonomous closed-loop tactile force regulation, leading to frequent object dropping, accidental crush damage, and amputee cognitive exhaustion.

Eleven Quantifiable Objectives:

1. **OBJ-01:** Synthesize deterministic 2-channel sEMG signals at 1000 Hz across 200 ms windows for 4 functional grip classes (Open Hand, Power Grip, Pinch Grip, Release).

2. **OBJ-02:** Compute 12 time-domain features (RMS, MAV, VAR, SSI, ZCR per channel, plus inter-channel ratio and difference) incorporating division-by-zero guards (\(N \le 1\)).
3. **OBJ-03:** Train and embed a constrained Random Forest classifier (35 trees, max depth 7) occupying < 200 KB storage.
4. **OBJ-04:** Profile host CPU inference latency dynamically using monotonic timers, verifying execution within a 10.0 ms budget.
5. **OBJ-05:** Implement closed-loop tactile force tracking toward a 50.0 N target with a 5.0 N contact threshold and \(\pm 3.0 ext{ N}\) deadband.
6. **OBJ-06:** Implement an autonomous shear slip reflex loop providing an instantaneous +18% (+12 N) effort boost to arrest slipping within 20 ms.
7. **OBJ-07:** Enforce a hard anti-crush safety ceiling of 75.0 N to protect fragile manipulated payloads.
8. **OBJ-08:** Expose FastAPI REST endpoints enforcing Pydantic validation (returning HTTP 422 for malformed payloads rather than HTTP 500 crashes).
9. **OBJ-09:** Develop a responsive High-DPI dashboard in React 19 / TypeScript with 3D kinematic hand visualization.
10. **OBJ-10:** Implement an automated continuous 27-second 5-stage demonstration state sequence.
11. **OBJ-11:** Maintain 100% pass rate across an automated regression test suite of 27 unit and integration tests.

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3. System Architecture & Methodology

The system establishes a four-tier hierarchical cyber-physical architecture:

- **Layer 1 (Sensory Simulation Tier):** Models the Flexor Digitorum Superficialis (Ch1) and Extensor Digitorum Communis (Ch2) muscles at \(f_s = 1000 ext{ Hz}\) across \(N = 200\) sample buffers with configurable baseline Gaussian noise \(\sigma_{base} \in [0.005, 0.20] ext{ V}\).
- **Layer 2 (Edge Intelligence Tier):** Extracts 12 guarded time-domain features and evaluates a 35-estimator Random Forest classifier, profiling microsecond host latency.
- **Layer 3 (Closed-Loop Tactile Control Tier):** Implements proportional force tracking (\(+1.5 ext{ N/cycle}\)), deadband stabilization (\([47.0, 53.0] ext{ N}\)), slip recovery boost (+18%), and anti-crush clamp (75.0 N).
- **Layer 4 (Telemetry Presentation Tier):** Renders High-DPI waveforms, live tactile force gauges, 3D hand kinematics, and continuous demo orchestration via React 19 and Vite.

Component	Technology / Algorithm	Key Technical Specification
Signal Synthesis	Deterministic Gaussian Envelopes	2 Channels, 1000 Hz, \(N=200\), 4 Grip Classes
Feature Extraction	Guarded Time-Domain Metrics	12 Features: RMS, MAV, VAR, SSI, ZCR, Ratio, Diff
ML Classification	Constrained Random Forest	35 Trees, Max Depth 7, Model Size 58 KB
Tactile Regulation	Closed-Loop Proportional Controller	Threshold: 5.0 N, Target: 50.0 N, Deadband: 3.0 N, Ceiling: 75.0 N
Anti-Slip Reflex	Autonomous Effort Boost	+18% Force Boost (+12.0 N) within 20 ms (1 cycle)
Backend API	FastAPI & Pydantic	7 REST Endpoints, HTTP 422 Error Handling, CORS Whitelist
Frontend Dashboard	React 19, TypeScript, Vite	Responsive High-DPI DPR Canvas, 3D Kinematics, 27s Demo Script

4. Empirical Results & Verification Evidence

The software prototype was rigorously evaluated using automated testing, ML evaluation on synthetic signals, and host CPU latency benchmarking:

1. **Automated Regression Suite: 27 out of 27 automated tests passing** in 4.06 seconds via `pytest backend/test_pipeline.py -v`. Coverage includes feature edge cases (\(N=0, 1, 5\)), input validation (HTTP 422), controller transitions, and reset idempotency.

2. **Production Frontend Build:** Successfully compiled via `npm run build` with zero TypeScript errors and zero bundling warnings (13 modules, 203.22 kB JS, 13.91 kB CSS, 669 ms build time).
3. **Synthetic Classification Accuracy: 100.0% holdout accuracy** (144/144 test samples) and **100.0% mean 5-fold cross-validation accuracy** with a **Macro-F1 score of 1.0000** across all four synthetic grip classes.
4. **Host Latency Benchmarking:** Measured mean host CPU inference latency of **1.46 ms ± 0.08 ms** across 50 warmed-up trials, well below the 10.0 ms real-time ceiling. Projected embedded ESP32-S3 latency is 3.1–5.3 ms.
5. **Tactile Controller Dynamics:** Verified stable force regulation at 49.8 N within the deadband window, instantaneous +18% anti-slip recovery boost, and strict 75.0 N safety ceiling clamping.

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5. Five-Phase Hardware Transition Roadmap

To transition from the software simulation prototype to physical embedded hardware, the team has outlined a structured 5-phase engineering roadmap:

- **Phase 1 (Months 1–2): TinyML Embedded C/C++ Porting:** Export Random Forest trees to static C header tables via `emlearn`; execute fixed-point feature extraction on an Espressif ESP32-S3 (240 MHz) with target latency < 5 ms.
- **Phase 2 (Months 3–4): Analog Front-End & Dry Electrodes:** Design custom 2-channel differential instrumentation amplifier PCB (TI INA333) with active bandpass filtering (20–450 Hz), 50 Hz notch filter, and socket-embedded Ag/AgCl dry electrodes.
- **Phase 3 (Months 5–6): 3D-Printed Mechanical Hand Fabrication:** Print anthropomorphic 5-digit chassis using carbon-fiber PETG and flexible TPU joints, actuated by coreless metal-gear micro-servos via high-tensile Dyneema tendons.
- **Phase 4 (Months 7–8): Tactile Sensing & Physical Reflex Closure:** Integrate Interlink 402 FSR pressure sensors and micro-vibration accelerometers into silicone fingertip pads; close the physical 50 Hz tactile servo loop.
- **Phase 5 (Months 9–12): Amputee Clinical Trials:** Conduct IRB-approved clinical usability testing with transradial amputees using standardized assessment protocols (Southampton Hand Assessment Procedure, Box and Blocks Test).

6. Project Milestone Schedule (Gantt Overview)

Milestone Phase	Key Deliverables	Target Duration	Status
Literature Survey & Baseline Audit	Research Matrix (35 papers), Code Audit Report	Weeks 1–3	COMPLETED
Algorithmic Implementation & Fixes	Guarded Features, RF Classifier, FSR Controller	Weeks 4–7	COMPLETED
API & Full-Stack Integration	FastAPI Endpoints, React 19 Canvas, 27s Demo	Weeks 8–10	COMPLETED
Automated Testing & Verification	27 Pytest Suite, Latency Benchmarking, Build	Weeks 11–12	COMPLETED
Documentation Sprint	Final Report (80+ pp), Synopsis, Midterm, Paper	Weeks 13–14	COMPLETED
Hardware Prototyping (Roadmap)	ESP32-S3 Firmware, AFE PCB, 3D Hand Chassis	Semester VIII	PLANNED

7. Key Academic References

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